

BAHAN_AJAR_Pertemuan04

BAHAN AJAR - PERTEMUAN 4 (S2)

JSON/CSV & Python

TP	AD, AP — Pengolahan Data Bervolume Besar
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A. FORMAT DATA — CSV

Struktur CSV

CSV = Comma-Separated Values. Data tabel di mana setiap baris adalah record dan kolom dipisahkan koma.

```
Nama,Kelas,Nilai,Status
Budi Santoso,X-A,85,Lulus
Citra Dewi,X-B,90,Lulus
Adi Pratama,X-A,78,Lulus
Dian Kurniawan,X-C,45,Remedial
```

Baris	Nama	Kelas	Nilai	Status
1 (header)	Nama	Kelas	Nilai	Status
2	Budi Santoso	X-A	85	Lulus
3	Citra Dewi	X-B	90	Lulus
4	Adi Pratama	X-A	78	Lulus
5	Dian Kurniawan	X-C	45	Remedial

Aturan CSV

Aturan	Contoh	Penjelasan
Delimiter koma	Budi, X-A, 85	Pemisah default
Quote jika ada koma	"Budi, S. T.", X-A, 85	Koma dalam data harus di-quote
Header opsional	Baris pertama	Jika ada, jadi nama kolom
Newline = baris baru	Enter	Setiap baris = 1 record

Kelebihan & Kekurangan CSV

Kelebihan	× Kekurangan
Format paling sederhana	Tidak ada tipe data (semua string)
Bisa dibuka Excel / Google Sheets	Tidak support nested data
Ukuran file kecil	Tidak ada standar encoding

Kelebihan	× Kekurangan
Standar universal — semua tool support	Koma dalam data butuh penanganan khusus

B. FORMAT DATA — JSON

Struktur JSON

JSON = JavaScript Object Notation. Format berbasis teks untuk pertukaran data — mirip dictionary Python.

Object (dict):

```
{
  "nama": "Budi Santoso",
  "kelas": "X-A",
  "nilai": 85,
  "lulus": true,
  "alamat": null
}
```

Array (list) of objects:

```
[
  {"nama": "Budi", "kelas": "X-A", "nilai": 85},
  {"nama": "Citra", "kelas": "X-B", "nilai": 90}
]
```

Nested (objek bersarang):

```
{
  "sekolah": "SMAN 6 Cimahi",
  "tahun": 2026,
  "siswa": [
    {"nama": "Budi", "nilai": {"uts": 85, "uas": 88}},
    {"nama": "Citra", "nilai": {"uts": 90, "uas": 92}}
  ]
}
```

Tipe Data JSON

JSON	Python	Contoh
string	str	"Budi"
number	int / float	85 , 3.14
boolean	bool	true / false
null	None	null
array	list	[1, 2, 3]
object	dict	{"key": "value"}

Kelebihan & Kekurangan JSON

Kelebihan	× Kekurangan
Support nested data	Ukuran lebih besar dari CSV
Tipe data eksplisit	Tidak bisa langsung dibuka Excel
Standar API web modern	Butuh parser untuk dibaca
Mudah dibaca manusia	Syntax lebih ketat dari CSV

C. PYTHON — MEMBACA & MENULIS FILE

Membuka File

```
# Mode membuka file
'r' # read (baca) – default
'w' # write (tuliskan) – overwrite
'a' # append (tambah)
'r+' # read + write

# Best practice: gunakan 'with' – auto close
with open('file.csv', 'r') as file:
    content = file.read()
```

1. Membaca CSV dengan `csv.reader`

```
import csv

with open('nilai_siswa.csv', 'r') as file:
    reader = csv.reader(file)
    header = next(reader) # ambil baris pertama (header)
    print('Header:', header)

    for row in reader:
        nama, kelas, nilai = row
        print(f'{nama} ({kelas}): {nilai}')
```

2. Membaca CSV dengan `csv.DictReader`

```
import csv

with open('nilai_siswa.csv', 'r') as file:
    reader = csv.DictReader(file)
    for row in reader:
        print(f'{row["Nama"]} ({row["Kelas"]}): {row["Nilai"]}')'
```

3. Menulis CSV dengan `csv.writer`

```
import csv

data = [
    ['Nama', 'Kelas', 'Nilai'],
    ['Budi', 'X-A', 85],
    ['Citra', 'X-B', 90],
]

with open('output.csv', 'w', newline='') as file:
    writer = csv.writer(file)
    writer.writerows(data)
```

4. Menulis CSV dengan `csv.DictWriter`

```
import csv

data = [
    {'Nama': 'Budi', 'Kelas': 'X-A', 'Nilai': 85},
    {'Nama': 'Citra', 'Kelas': 'X-B', 'Nilai': 90},
]

with open('output.csv', 'w', newline='') as file:
    fieldnames = ['Nama', 'Kelas', 'Nilai']
    writer = csv.DictWriter(file, fieldnames=fieldnames)
    writer.writeheader()
    writer.writerows(data)
```

5. Membaca JSON

```
import json

with open('data_siswa.json', 'r') as file:
    data = json.load(file)

print(type(data)) # list atau dict
print(json.dumps(data, indent=2)) # cetak rapi
```

6. Menulis JSON

```
import json

data = [
    {'nama': 'Budi', 'kelas': 'X-A', 'nilai': 85},
    {'nama': 'Citra', 'kelas': 'X-B', 'nilai': 90},
]

with open('output.json', 'w') as file:
    json.dump(data, file, indent=2)
```

D. STUDI KASUS: KONVERSI CSV ↔ JSON

CSV → JSON

```

import csv
import json

csv_file = 'nilai_siswa.csv'
json_file = 'nilai_siswa.json'

data = []
with open(csv_file, 'r') as f:
    reader = csv.DictReader(f)
    for row in reader:
        row['Nilai'] = int(row['Nilai']) # konversi string ke int
        data.append(row)

with open(json_file, 'w') as f:
    json.dump(data, f, indent=2)

print(f'Konversi selesai: {csv_file} → {json_file}')

```

JSON → CSV

```

import csv
import json

json_file = 'nilai_siswa.json'
csv_file = 'nilai_siswa_dari_json.csv'

with open(json_file, 'r') as f:
    data = json.load(f)

with open(csv_file, 'w', newline='') as f:
    fieldnames = data[0].keys()
    writer = csv.DictWriter(f, fieldnames=fieldnames)
    writer.writeheader()
    writer.writerows(data)

print(f'Konversi selesai: {json_file} → {csv_file}')

```

E. RANGKUMAN

Operasi	CSV	JSON
Struktur	Tabel (baris × kolom)	Key-value / nested
Module	csv	json
Baca file	csv.reader(), csv.DictReader()	json.load()
Tulis file	csv.writer(), csv.DictWriter()	json.dump()
Python → String	—	json.dumps()
String → Python	—	json.loads()
Kelebihan	Sederhana, ringan, Excel	Nested, tipe data, API

